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Written Assignment 1

Q1. Write a C program to merge two one dimensional arrays.

A1.
```
#include <stdio.h>

void main() {
    int c1= 3, c2=4;
    int arr1[] = {2, 1, 4};
    int arr2[] = {5, 7, 9, 3};

    int ans[c1+c2];
    for(int i=0; i<c1; i++) {
        ans[i] =arr1[i];
    }
    for (int i= 0; i<c2; i++) {
        ans [c1+i] = arr2[i];
    }
    printf("Merged Array: ");
    for(int i=0; i< c1+c2; i++){
        printf("%d ", ans[i]);
    }
}
```

Q2. Explain in detail multiple stack with diagrams and write a C program for two way stack.

A2. A multiple stack is a concept where more than one stack is implemented using a single array.
It allows us to dynamically allocate the size of each of the sub arrays ~~each~~ ~~A~~
There are 2 ways to implement multiple stack. 2 way stack & Multiple stack.

2 way stack has elements from both ends of the array and each of the stack can dynamically resize each other depending on their needs. ~~They~~ One array is indexed from 0 to the max size and the other is from max size to zero. ~~This~~ All overlaps are to be taken into account.

1	4	5	6	7	8	3	2	4	5

Array 1 Array 2

This shows how the size of each array can be dynamically allocated.

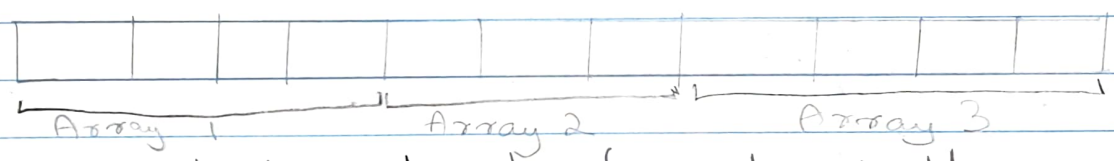
Pushing to array 1 - moves its end one space towards the right and ~~a~~ ~~gets the number to~~ store the number
Popping array 1 - moves its end one space towards the start of the array.

~~Popp~~Pushing to array 2 - Moves its end one space towards the start and stores the number.

Popping array 2 - Moves its end one space towards the end of the array.

Multiple stack.

It divides the array into 3 or more parts and store the start and ~~number of~~ ending indexes of the ~~arrays~~ individual sub arrays.



The start and end of each of the ~~arrays~~ indexes of each of the arrays is stored.

The start values remain constant of all and are used when the arrays are displayed.

The ends are ~~added~~ incremented and decremented as and when push and pop are called respectively.

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C program for two way stack

```
#include <stdio.h>
void pushs1 (int s[], int *c1, int *c2, int num) {
    if ((*c1) <= (*c2)) {
        s[*c1] = num;
        (*c1)++;
    } else {
        printf("Error\n");
    }
}

void pushs2 (int s[], int *c1, int *c2, int num) {
    if ((*c1) <= (*c2)) {
        s[*c2] = num;
        (*c2)--;
    } else {
        printf("Error\n");
    }
}

int popS1 (int s[], int *c) {
    ~~int val = s[*c];~~
    ~~(*c)--;~~
    ~~return val;~~
    if ((*c) > 0) {
        int val = s[*c]; (*c)--; return val;
    } else {
        printf("Error\n");
    }
```

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```
int popS2(int s[], int *c) {
    ~~int val = s[*c]; if((*c)<size-1){~~
    ~~(*c)++;~~ int val = s[*c]; (*c)++; return val;
    ~~return val;~~ } else {
        printf("Error\n");
    }

void display(int s[], int s1, int s2, int size){
    for(int i=0; i<s1; i++){
        printf("%d ", s[i]);
    }
    for(int i = size-1; i>s2; i--){
        printf("%d ", s[i]);
    }
}
void main(){
    int size = 5;
    int s[size];
    int s1t = 0, s2t = size-1;
    pushs1(s, &s1t, &s2t, 10);
    pushs1(s, &s1t, &s2t, 11);
    pushs1(s, &s1t, &s2t, 12);
    push s2(s, &s1t, &s2t, 13);
    pops1(s, &s1t);
    pop s2(s, &s2t, size);
    display(s, s1t, s2t, size);
}
```

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Q3. Describe multiple queues with suitable digrams.

Ans. Multiple queus mean two or more queues are maintained simultaneously, elements are distributed between them and they all are separate independent ~~ob~~ instances of the main struct. Each queue operates indepently with common insert, delete and display functions, but the queue ~~names are passed~~ to be modified pointer is passed as function argments to the function.

Queue1 -

Queue 2 -

Queue 3

Each queue can have different sizes ~~as an~~ depending on the developers need.
Each queue has its array, front and rear stored.

There are 3 standard functions of enqueue, dequeue and display.
It ~~moves~~ moves start right when dequeue is called and end to the right when enqueue is called.
Display iterates from start to ending indexes of the queue.